BUSINESS REQUIREMENTS DOCUMENT

ECommerce Reference

Feature Summaries
Prepared by Kolosal AI

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Product Detail Page

Prepared by Kolosal AI

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1. Frontend

1.1 Overview

The Product Detail View is the central page where customers can view specific product information, select their preferences, and make purchasing decisions. The page is designed to be user-friendly, supporting multiple languages and regional sizing, while providing seamless navigation back to previous search results.

1.2 Step-by-Step Flow

1. Loading: When a user selects a product, the system fetches the product details from the database. A loading indicator is shown until the information is ready.
2. Displaying Information: Once loaded, the page displays the product title, customer ratings, and pricing. If a product is on sale, the original price is shown with a strike-through alongside the discounted price.
3. Customization: Users can select their preferred color and regional size (US, UK, or Asia). Changing the region automatically updates the size options to match that market.
4. Navigation: A "Back" button allows users to return to the previous page while preserving their previous search filters, sorting preferences, and scroll position.
5. Purchasing & Engagement:

* Buy Now: A direct purchase button is available, which automatically adjusts its label (e.g., "Buy Now" vs. "Acheter Maintenant") based on the user's language settings.
* Add to Cart: Users can add the item to their shopping cart.
* Wishlist: Users can save items to their wishlist. If a user is not logged in, they will be prompted to sign in first.
* Share: A share icon allows users to easily share the product link using their device's native sharing tools.

1.3 Key Concepts

- Dynamic Pricing: A display logic that automatically highlights discounts by showing the original price crossed out next to the new price.
- Regional Localization: The ability for the interface to adapt to different markets by changing the language of buttons and adjusting size options based on the selected region.
- State Preservation: A feature that ensures the user's browsing context (filters, scroll position) remains intact when navigating back to a previous page.
- Authentication Gate: A security measure that ensures users are logged in before performing actions like adding items to a personal wishlist.

1.4 Business Value

- Improved User Experience: By retaining search filters and scroll positions, customers can browse more efficiently without losing their place.
- Global Readiness: Multi-language support and regional size selectors allow the platform to serve international customers seamlessly.
- Increased Conversion: Clear pricing displays and direct "Buy Now" options reduce friction in the purchasing process, while wishlist functionality encourages repeat visits and long-term engagement.

2. Logic

2.1 Overview

This system manages how product information, customer reviews, and personalized recommendations are delivered to the website. It is designed to ensure that product pages load quickly by pre-calculating complex data (like review ratings and discounts) in the background, rather than forcing the system to perform heavy calculations while a customer is waiting for a page to load.

2.2 Step-by-Step Flow

1. Initial Page Load: When a customer views a product, the system simultaneously fetches the main product details (images, descriptions, and base pricing) and the review summary.
2. Discount Application: Before showing the price, the system checks for any active promotions and automatically calculates the final discounted price.
3. Review Display: The system retrieves the pre-calculated average rating and total review count, along with the list of individual reviews, ensuring the review section populates instantly.
4. Variant Selection: If a customer selects a specific option (like a different size or color), the system performs a "lightweight" lookup to update only the stock status and price for that specific choice, avoiding a full page refresh.
5. Recommendation Loading: The system identifies relevant products to suggest to the customer, pulling in their titles, prices, and ratings so they can be displayed immediately without extra steps.
6. Background Updates: Whenever a new review is submitted, the system automatically recalculates the product's average rating and total count in the background, keeping the data fresh for the next visitor.

2.3 Key Concepts

- Pre-computation: The practice of calculating values (like average ratings) when data changes, rather than when a user requests it. This keeps the website fast.
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Lightweight Lookups: Targeted data requests that fetch only the specific information needed (like a price change for a specific size) instead of reloading the entire product page.

- Variant: A specific version of a product, such as a different color or size, which may have its own unique stock level or price.
- Aggregate Figures: Summary data, such as the total number of reviews or the average star rating, which provides a quick snapshot of customer sentiment.

2.4 Business Value

- Improved Performance: By pre-calculating ratings and using lightweight lookups for variants, the website remains responsive and fast, which is proven to increase customer satisfaction and conversion rates.
- Data Accuracy: Automated background updates ensure that customers always see the most current pricing and the most accurate review scores without manual intervention.
- Seamless User Experience: Customers can interact with product options (like selecting a size) and see updated information instantly, creating a smooth, professional shopping experience.

3. Data

3.1 Overview

This feature manages how product information, customer reviews, and pricing are displayed and updated on our platform. It ensures that when a customer views a product, they see accurate details—including images, available color options, current stock, and active discounts—while also providing a smooth experience for submitting and viewing product reviews.

3.2 Step-by-Step Flow

1. Product Loading: When a user views a product page, the system gathers core details (title, description, base price), active promotions, available images, and product variants (like colors and stock levels) to display the full product profile.
2. Dynamic Updates: If a user selects a specific product variant (e.g., a different color), the system instantly fetches the updated stock and pricing information for that specific choice.
3. Review Submission: When a customer submits a review, the system saves it to the database. It enforces a rule that prevents the same user from submitting multiple reviews for the same product.
4. Rating Recalculation: Immediately after a new review is saved, the system automatically updates the product's overall average rating and total review count.
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Review Display: When a user views the reviews section, the system retrieves the pre-calculated average rating and count alongside the list of individual customer comments.

3.3 Key Concepts

- Product Variant: A specific version of a product, such as a different color, which may have its own unique stock level or price.
- Promotion: A temporary discount applied to a product that the system automatically detects and calculates during the page load.
- Rating Metrics: The pre-calculated average score and total number of reviews for a product, which are updated in real-time to ensure the display is always current.
- Data Integrity: Rules in place to ensure that each customer can only leave one review per product, preventing duplicate feedback.

3.4 Business Value

- Performance: By pre-calculating ratings, the system avoids heavy processing when a user loads a product page, ensuring the site remains fast even as the number of reviews grows.
- Accuracy: Customers always see the most up-to-date pricing, stock availability, and product feedback, which builds trust and reduces purchase friction.
- Scalability: The system is designed to handle large volumes of product data and reviews efficiently, ensuring that the shopping experience remains consistent as our catalog and customer base expand.

BUSINESS REQUIREMENTS DOCUMENT

Smart Suggestion

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1. Frontend

1.1 Overview

The search feature provides a user-friendly way to find products, brands, or categories across the platform. It is designed to be intuitive and efficient, offering helpful shortcuts like recent search history and real-time suggestions as the user types. The system also includes built-in safeguards to ensure search queries remain within a reasonable length.

1.2 Step-by-Step Flow

1. Engagement: When a user clicks or taps the search box, the system immediately displays their recent search history to help them quickly find items they have looked for previously.
2. Typing: As the user begins to type, the system provides real-time, relevant suggestions to speed up the search process.
3. Validation: The system monitors the input length. If the user exceeds 100 characters, the system will prevent further input or flag the error to ensure the search remains effective.
4. Refinement: If the user makes a mistake or wants to start over, they can use the "clear" icon to wipe the current text or the "clear all" button to remove their entire search history.
5. Submission: The user can trigger the search by pressing the "Enter" key. The system then navigates to a results page, keeping the search term visible so the user knows exactly what they are viewing.
6. Dismissal: If the user decides not to search, they can press the "Escape" key to hide the suggestions and close the search interface.

1.3 Key Concepts

- Auto-suggestions: A feature that predicts what the user is looking for based on their current input, helping them find items faster.
- Recent Searches: A personalized list of previous queries that appears when the search box is activated, allowing for quick re-discovery of past interests.
- Input Validation: A background check that ensures the search query is not too long (capped at 100 characters) to maintain system performance and search accuracy.
- Search History Management: Tools that allow users to maintain their privacy and keep their search interface clean by clearing individual or all past search entries.

1.4 Business Value

- Improved User Experience: By providing suggestions and recent history, we reduce the effort required for users to find what they need, leading to higher satisfaction.
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Increased Conversion: Faster, more accurate search results help users reach product pages more quickly, which directly supports sales goals.

- Data Integrity: By enforcing character limits, we ensure that search queries are optimized for our database, preventing system errors and ensuring relevant results are returned every time.

BUSINESS REQUIREMENTS DOCUMENT

Login with username and password

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1. Frontend

1.1 Overview

The login feature provides a secure and user-friendly way for users to access their accounts. It supports standard email-based authentication with built-in validation to ensure data accuracy. The interface is designed to be intuitive, offering features like password visibility toggling and clear visual feedback during the login process to improve the overall user experience.

1.2 Step-by-Step Flow

1. Input Credentials: The user enters their email address and password into the respective fields. The system automatically cleans up any accidental spaces at the beginning or end of the input.
2. Validation: As the user types, the system checks that the email format is valid and that the password field is not empty.
3. Submission: Once both fields are correctly filled, the "Login" button becomes active. The user clicks the button to submit their credentials.
4. Processing: The system disables the button and displays a loading indicator to prevent multiple accidental submissions while it verifies the credentials.
5. Outcome:

* Success: The system processes the login and redirects the user to their dashboard.

* Failure: If the credentials do not match, the system displays a clear error message, allowing the user to correct their information and try again.

1.3 Key Concepts

- Masked Input: A security feature where the password appears as dots or asterisks by default to prevent others from seeing it while the user types.
- Visibility Toggle: A control that allows the user to temporarily reveal their password in plain text to verify they typed it correctly.
- Form Validation: Real-time checks that ensure the user provides data in the correct format (e.g., a valid email address) before the system attempts to process the login.
- Loading State: A visual indicator that informs the user that the system is currently working on their request, which helps manage expectations and prevents duplicate clicks.

1.4 Business Value

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Enhanced Security: By masking passwords and providing a visibility toggle, we protect user privacy while reducing frustration caused by typing errors.

- Improved Data Quality: Automatic removal of leading and trailing spaces and real-time format validation ensure that only clean, valid data enters our system.
- Reduced Support Overhead: Clear, immediate error messaging helps users resolve their own login issues without needing to contact customer support.
- Optimized Performance: Disabling the button during the loading process prevents duplicate requests, which reduces unnecessary server load and potential errors.

2. Logic

2.1 Overview

This feature enables a seamless "Sign in with Google" experience for our users. It allows new and returning users to access our platform using their existing Google credentials, eliminating the need to create and remember a separate password for our service.

2.2 Step-by-Step Flow

1. The system receives the user's profile information directly from Google after they successfully log in.
2. The system checks our database to see if the user already has an account associated with that email address.
3. If the user is already registered, the system grants them access to their existing account.
4. If the user is new, the system automatically creates a new account profile for them using their Google name and email address.
5. Once the account is verified or created, the system generates a secure session token, allowing the user to stay logged in for the next 24 hours.

2.3 Key Concepts

- Google OAuth: A secure industry-standard method that allows users to sign into our application using their Google account without sharing their Google password with us.
- Session Token: A temporary digital "key" that proves the user is logged in, allowing them to navigate our site securely without having to re-enter their credentials repeatedly.
- User Synchronization: The process of matching incoming Google profile data with our internal records to ensure the user is correctly identified.

2.4 Business Value

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Improved User Experience: Reduces friction during sign-up and login, which typically leads to higher conversion rates and fewer abandoned registrations.

- Enhanced Security: By leveraging Google's robust authentication infrastructure, we reduce the risks associated with managing user passwords and potential credential theft.
- Operational Efficiency: Automates the account creation process, removing the need for manual verification or complex onboarding steps for new users.

3. Data

3.1 Overview

This feature manages the user login process, ensuring that only authorized users can access the system while protecting accounts from unauthorized access attempts. It includes a security mechanism that tracks failed login attempts and automatically locks accounts if they are targeted by repeated incorrect password entries.

3.2 Step-by-Step Flow

1. The user submits their username and password through the login portal.
2. The system checks if the account exists and verifies that it is not currently locked.
3. The system compares the provided password against the stored secure password record.
4. If the password is correct, the system clears any previous failed login counts and issues a secure access token to grant the user entry.
5. If the password is incorrect, the system records the failed attempt.
6. If the number of failed attempts reaches five, the system automatically locks the account to prevent further access.
7. The system returns a success message with an access token for valid logins, or an error message for failed attempts.

3.3 Key Concepts

- Authentication: The process of verifying a user's identity by matching their credentials against our records.
- Account Lockout: A security measure that disables an account after five consecutive failed login attempts to protect against unauthorized access.
- Access Token: A digital key provided to the user after a successful login, which allows them to interact with the system securely.
- Failed Login Counter: A background tracker that monitors how many times a user has entered an incorrect password.

3.4 Business Value

This feature provides a robust security layer for our application. By automatically locking accounts after repeated failed attempts, we significantly reduce the risk of "brute-force" attacks where unauthorized parties try to guess passwords. Simultaneously, the system ensures a smooth experience for legitimate users by automatically resetting their security status upon a successful login, maintaining a balance between high security and user convenience.

BUSINESS REQUIREMENTS DOCUMENT

Cart

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1. Frontend

1.1 Overview

The shopping cart feature provides a user-friendly interface for customers to manage their selected items before completing a purchase. It allows users to view product details, adjust quantities, remove items, and see real-time price updates. The system also includes built-in safeguards to handle out-of-stock items and ensure the checkout process only proceeds when the cart contains valid products.

1.2 Step-by-Step Flow

1. View Cart: The system displays a list of items, showing the product image, name, selected options, price, current quantity, and the calculated subtotal for each item.
2. Adjust Quantity: Users can increase or decrease the number of items. If a user decreases the quantity to zero, the system automatically removes the item from the cart.
3. Automatic Updates: Whenever a quantity is changed, the system instantly recalculates the item subtotal and the overall cart total.
4. Remove Items: Users have the option to manually remove any item from their cart at any time.
5. Checkout: Once the user is ready, they can click the "Check Out" button to be redirected to the payment and shipping page.

1.3 Key Concepts

- Subtotal: The calculated cost for a specific product based on its unit price multiplied by the selected quantity.
- Total Price: The combined cost of all items currently in the shopping cart.
- Availability Validation: A safety feature that identifies out-of-stock items; these items have their quantity controls disabled to prevent users from attempting to purchase unavailable inventory.
- Checkout Readiness: A logic check that ensures the "Check Out" button is only active when there is at least one item in the cart, preventing empty transactions.

1.4 Business Value

- Improved User Experience: Real-time price updates and intuitive quantity controls reduce friction, making it easier for customers to manage their orders.
- Inventory Accuracy: By disabling controls for out-of-stock items, the system prevents customers from ordering products that are not available, reducing post-purchase support issues.
- Increased Conversion: By streamlining the path to checkout and providing clear feedback on cart totals, the feature encourages users to complete their purchases efficiently.